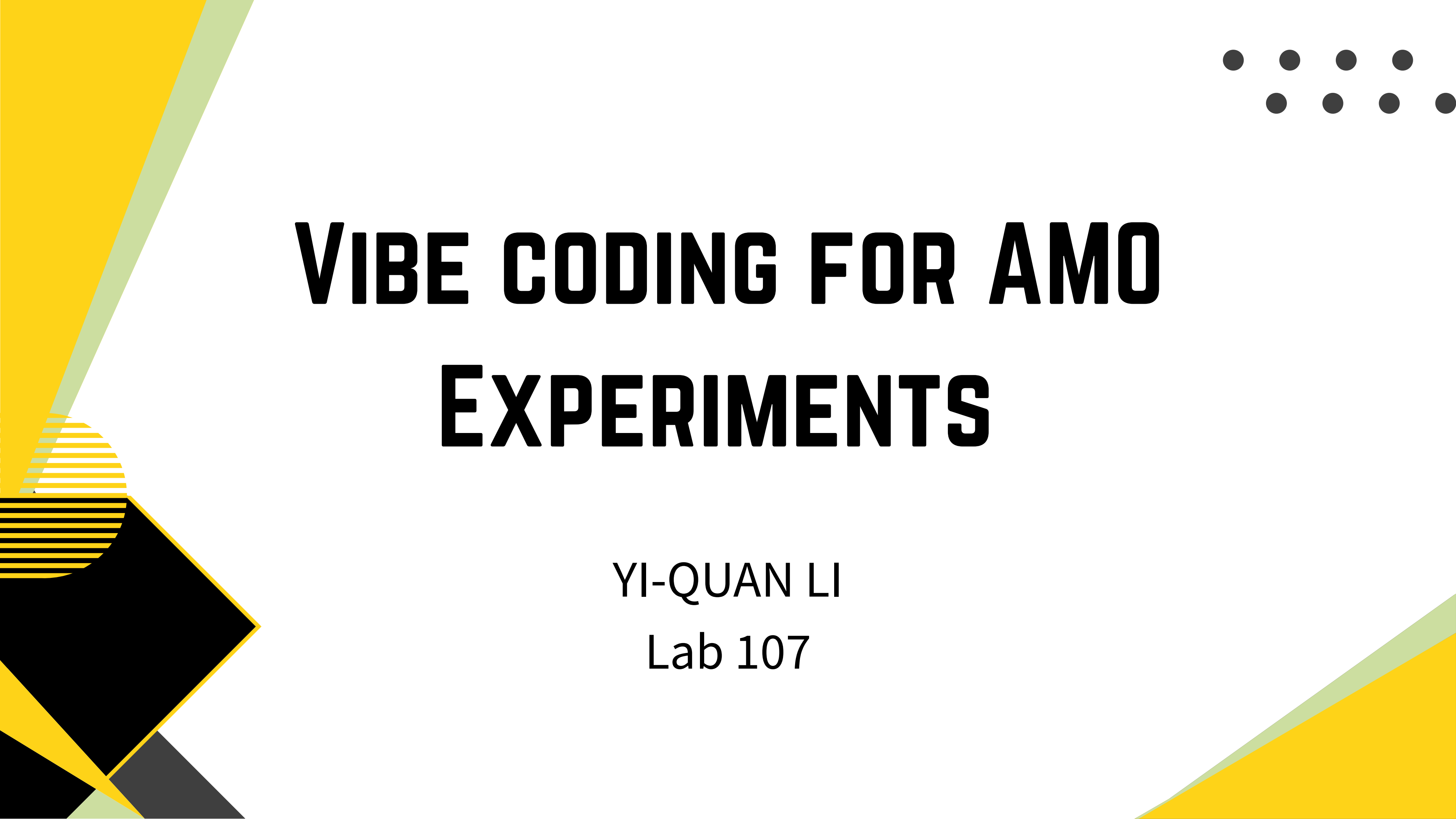




VIBE CODING FOR AMO EXPERIMENTS

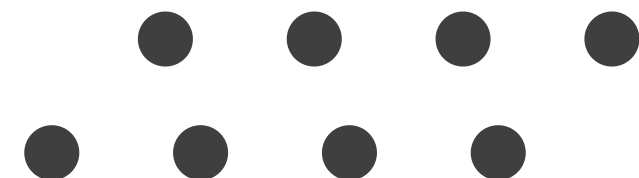
YI-QUAN LI
Lab 107



OUTLINE

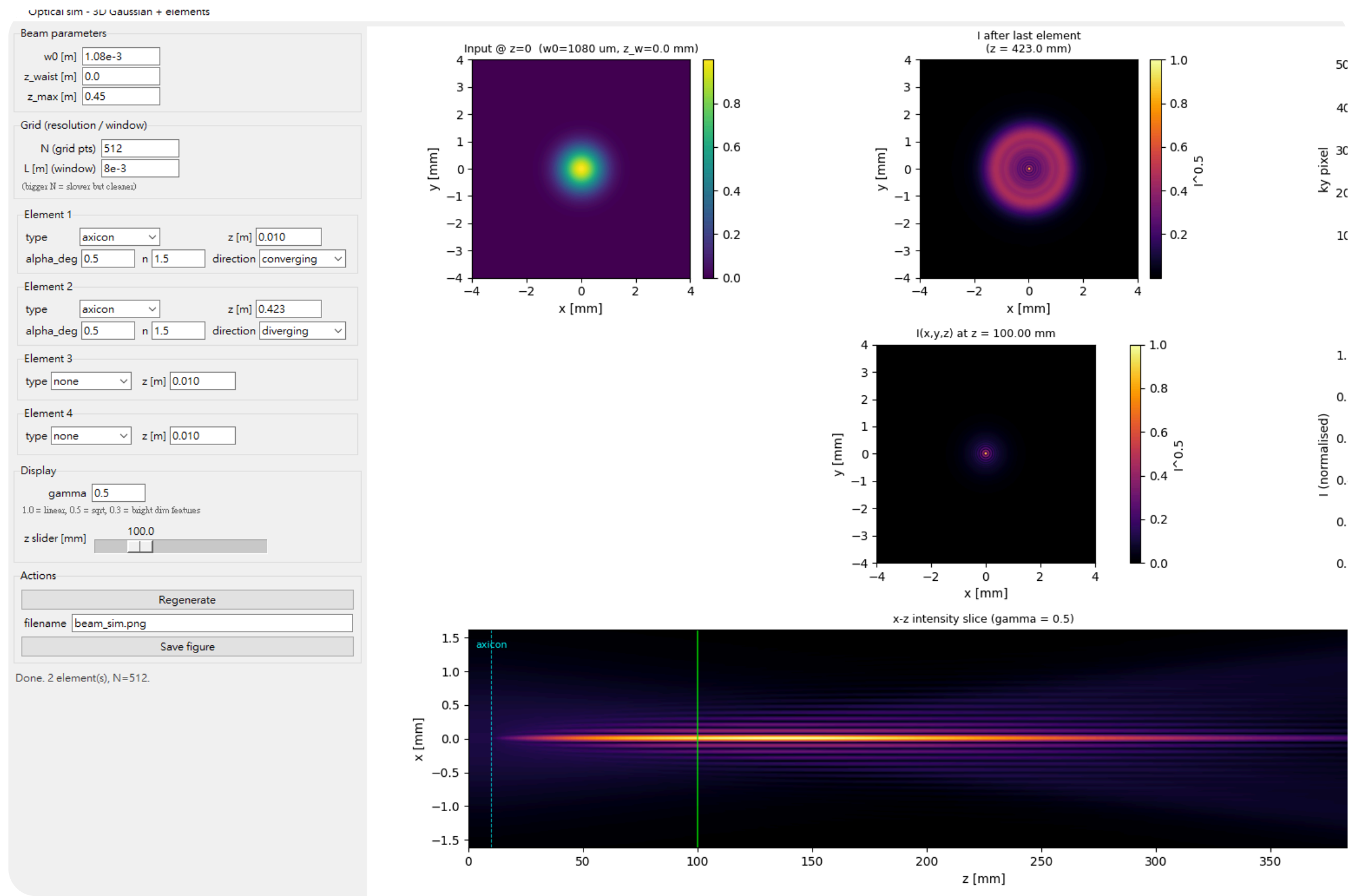


- **Optical simulator**
 - motivation
 - workflow
 - result
 - check and evaluation
- **Bluetooth Air-quality sensor**
 - motivation
 - workflow
 - result
 - check and evaluation



AI-GENERATED RAY TRACING GUI

- Insert optical components in 1D space
- Calculate and display the spatial profile at arbitrary location

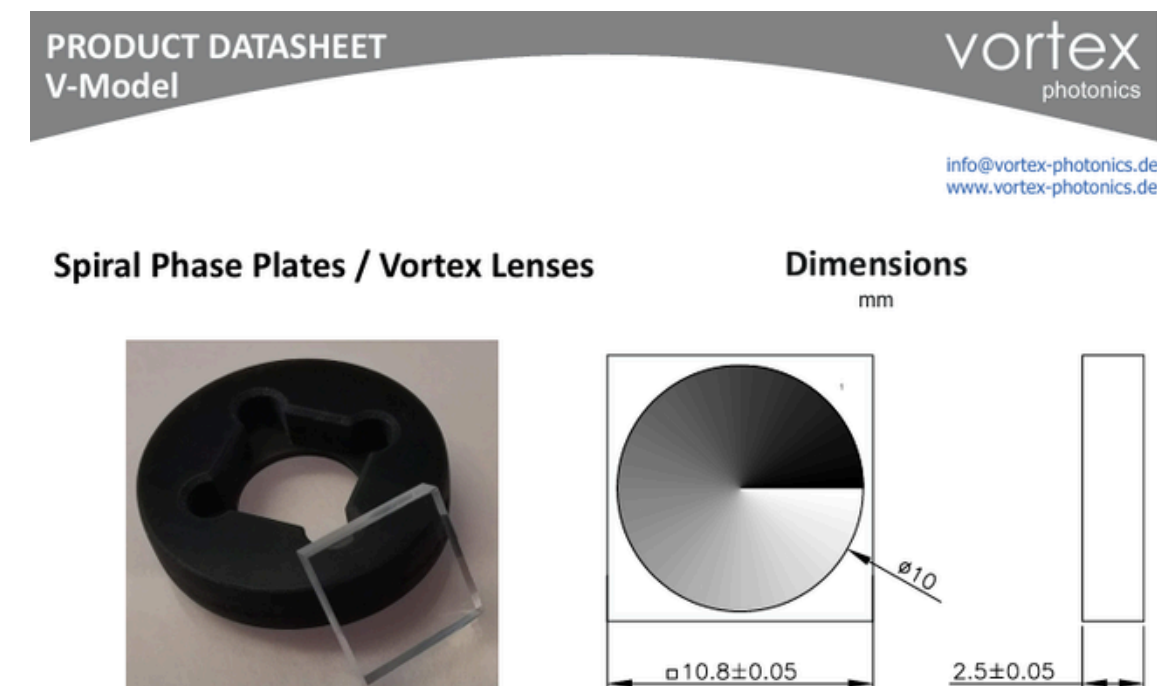
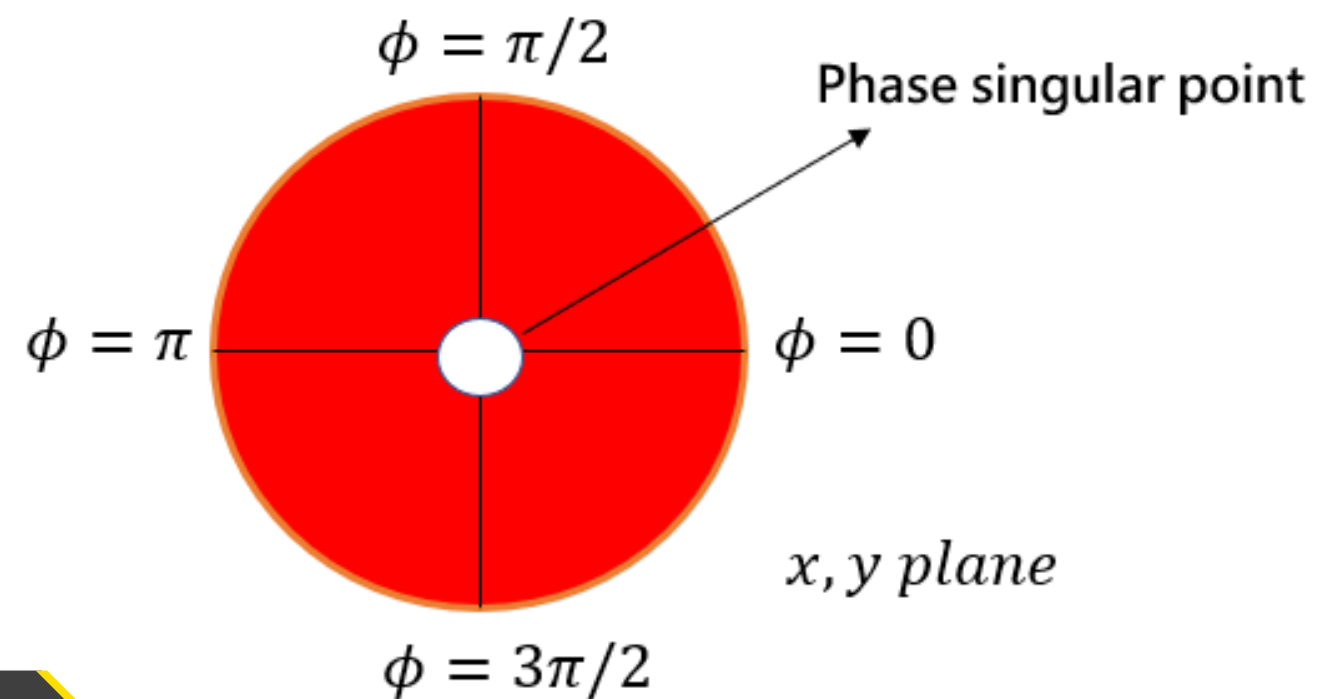


MOTIVATION

We want to generate a hollow beam for our experiment.

$$E(x, y) \rightarrow E(x, y)e^{il\phi}$$

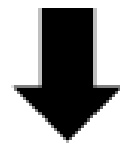
The most efficient way is to buy a spiral phase plate



What l should we choose or how to set it?

WORKING FLOW

- Ask **chat GPT** if there is an existing laser simulator available



- ChatGPT : I recommend you use python code to build one...



- Q: Based on the desired far-field beam profile, design a proper phase mask.



Claude



```
Python
import numpy as np
import matplotlib.pyplot as plt

# -----
# Parameters
# -----
N = 1024
L = 6e-3
w0 = 0.8e-3
ell = 1

x = np.linspace(-L/2, L/2, N, endpoint=False)
y = np.linspace(-L/2, L/2, N, endpoint=False)
X, Y = np.meshgrid(x, y)

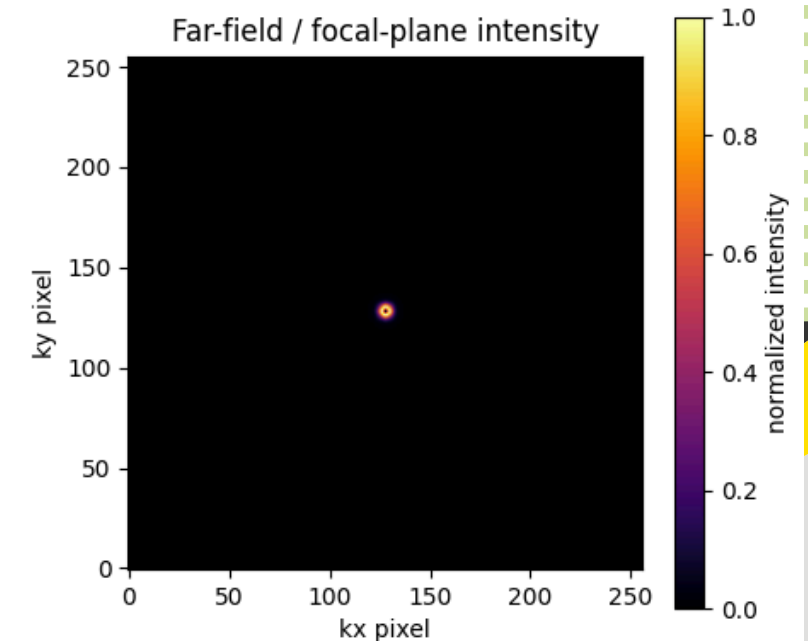
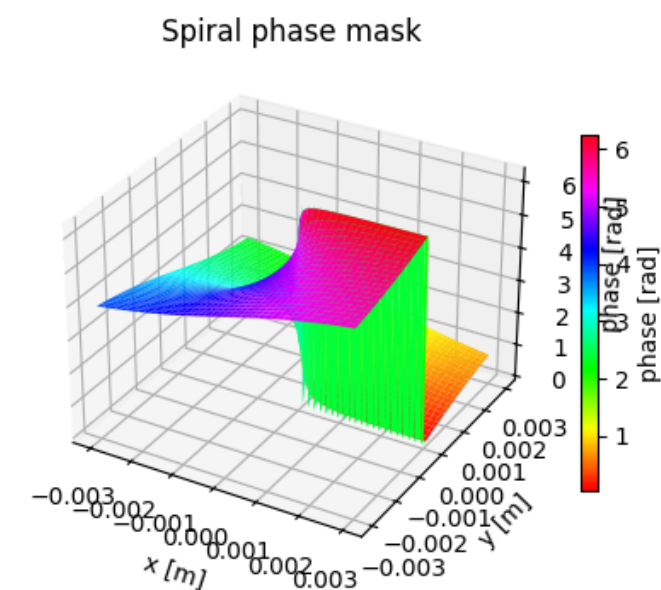
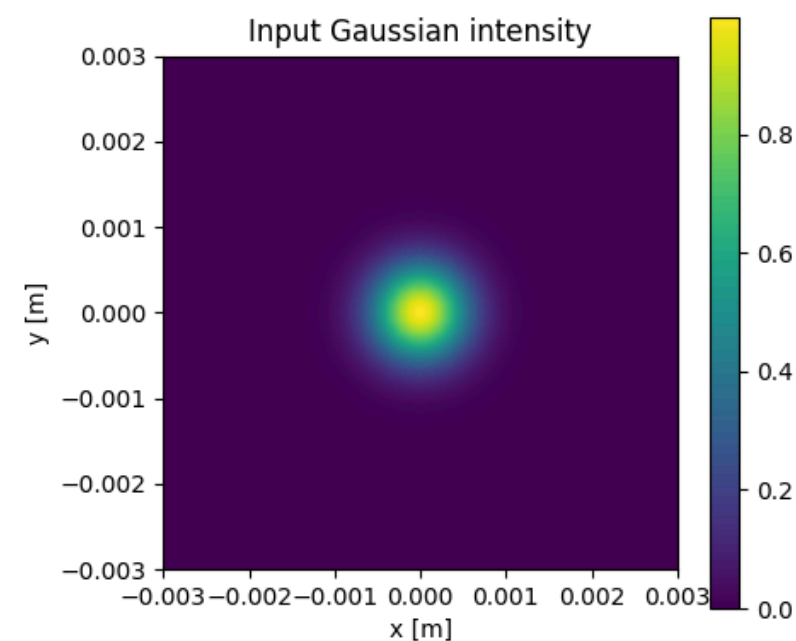
R = np.sqrt(X**2 + Y**2)
Theta = np.arctan2(Y, X)

# Gaussian input
E_in = np.exp(-R**2 / w0**2)

# Spiral phase mask
phase = ell * Theta

# Field after phase modulator
E_after = E_in * np.exp(1j * phase)

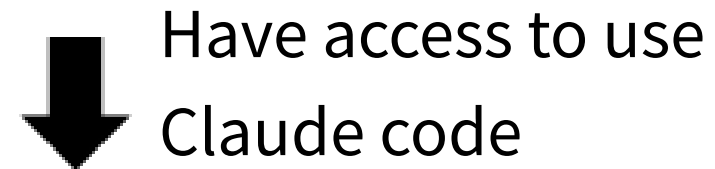
# Fourier plane / far field
E_far = np.fft.fftshift(
    np.fft.fft2(
        np.fft.ifftshift(E_after)
    )
)
```



WORKFLOW



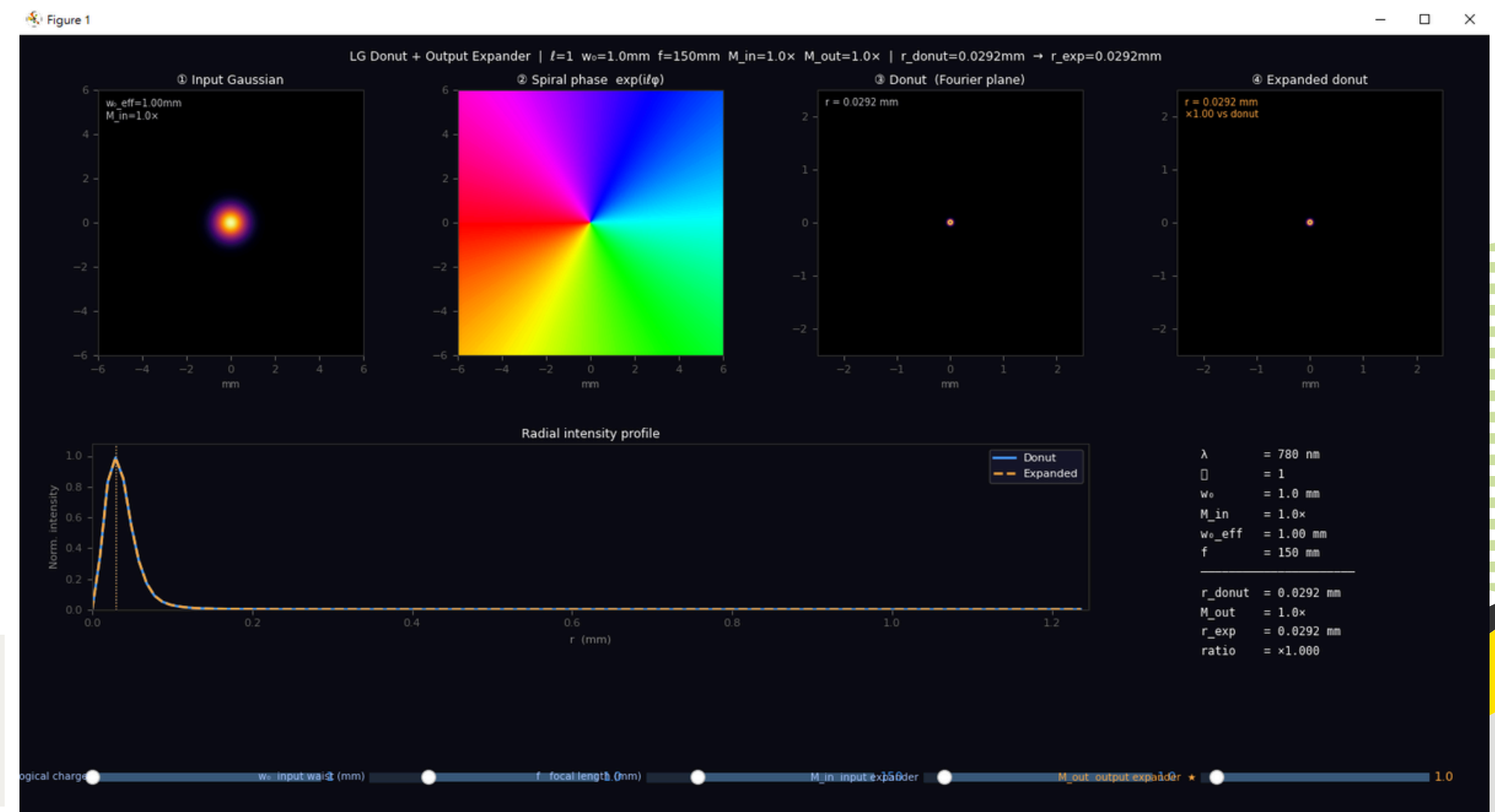
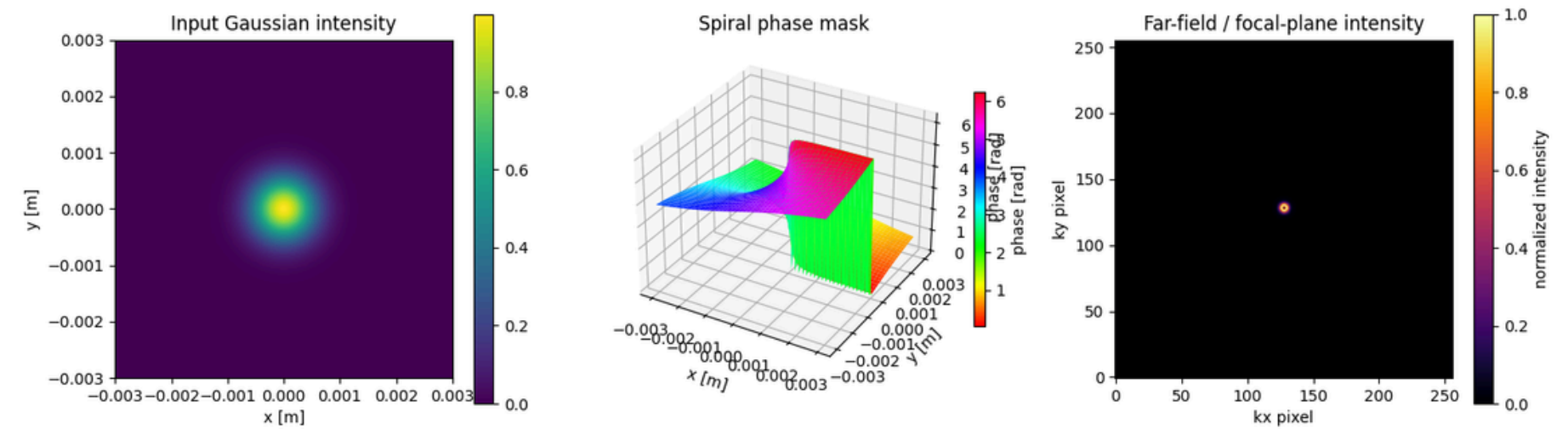
- Ask GPT about it idea and comprehend the theory behind it.



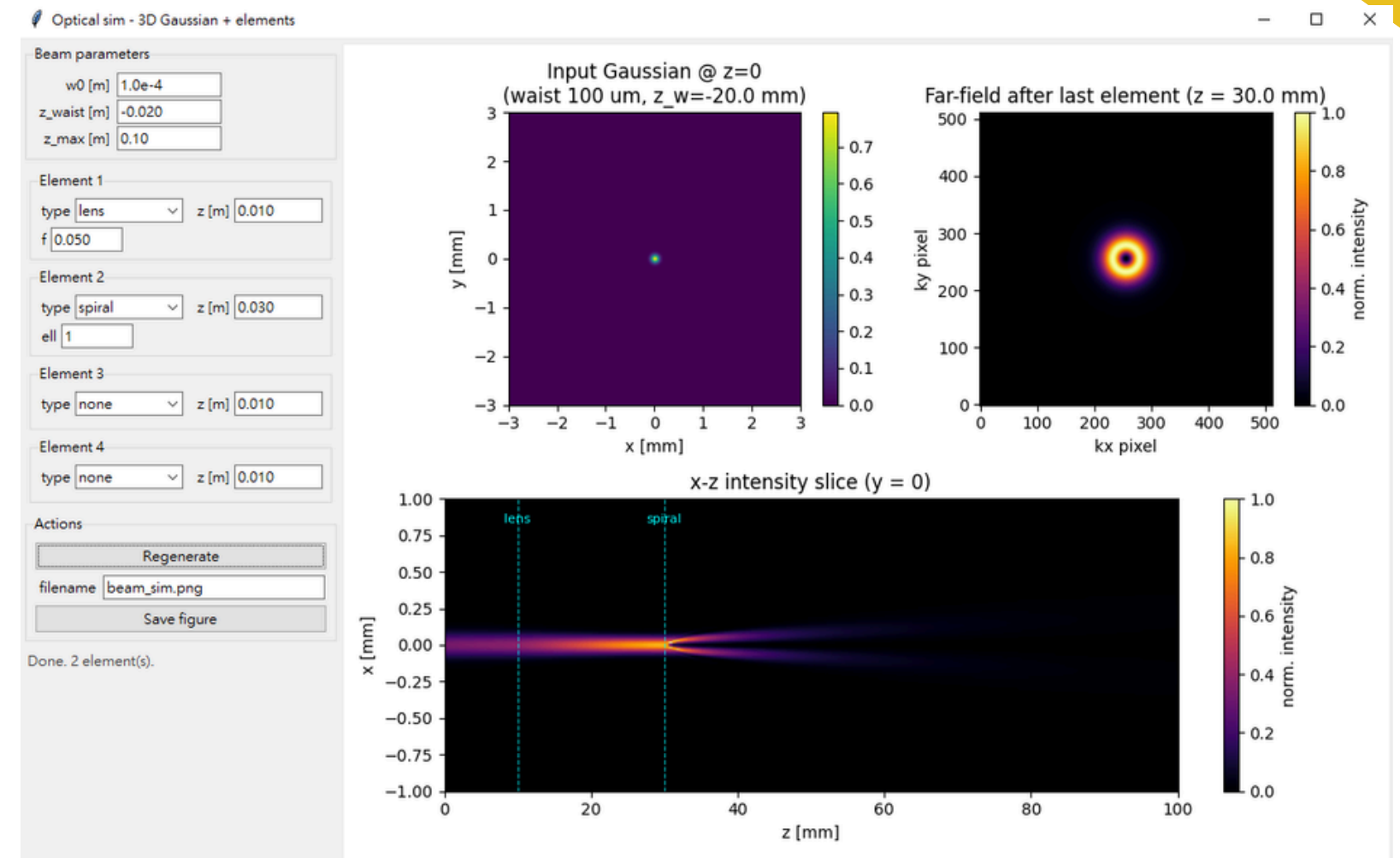
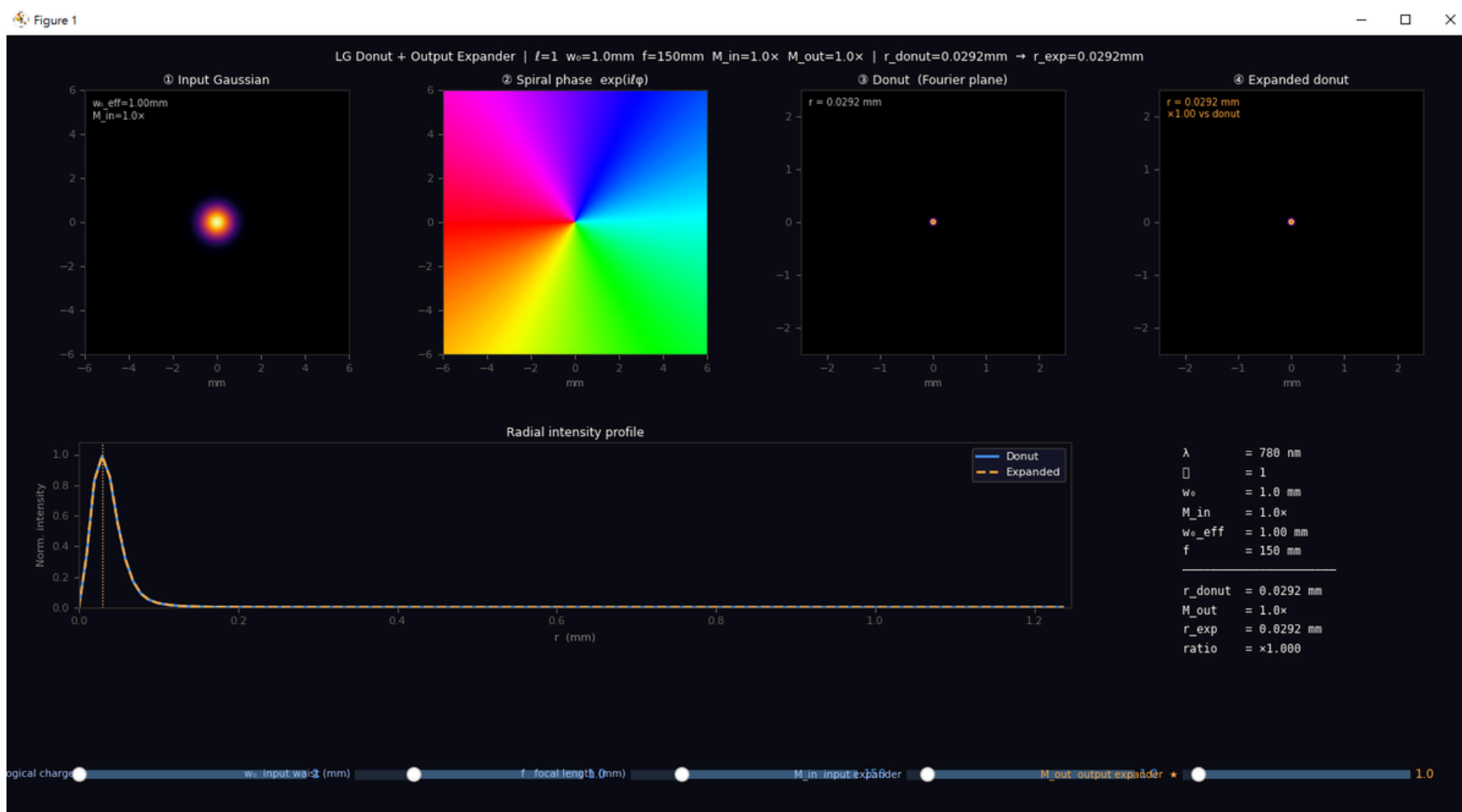
- Ask Claude to do the same thing.



- Fully functioning GUI



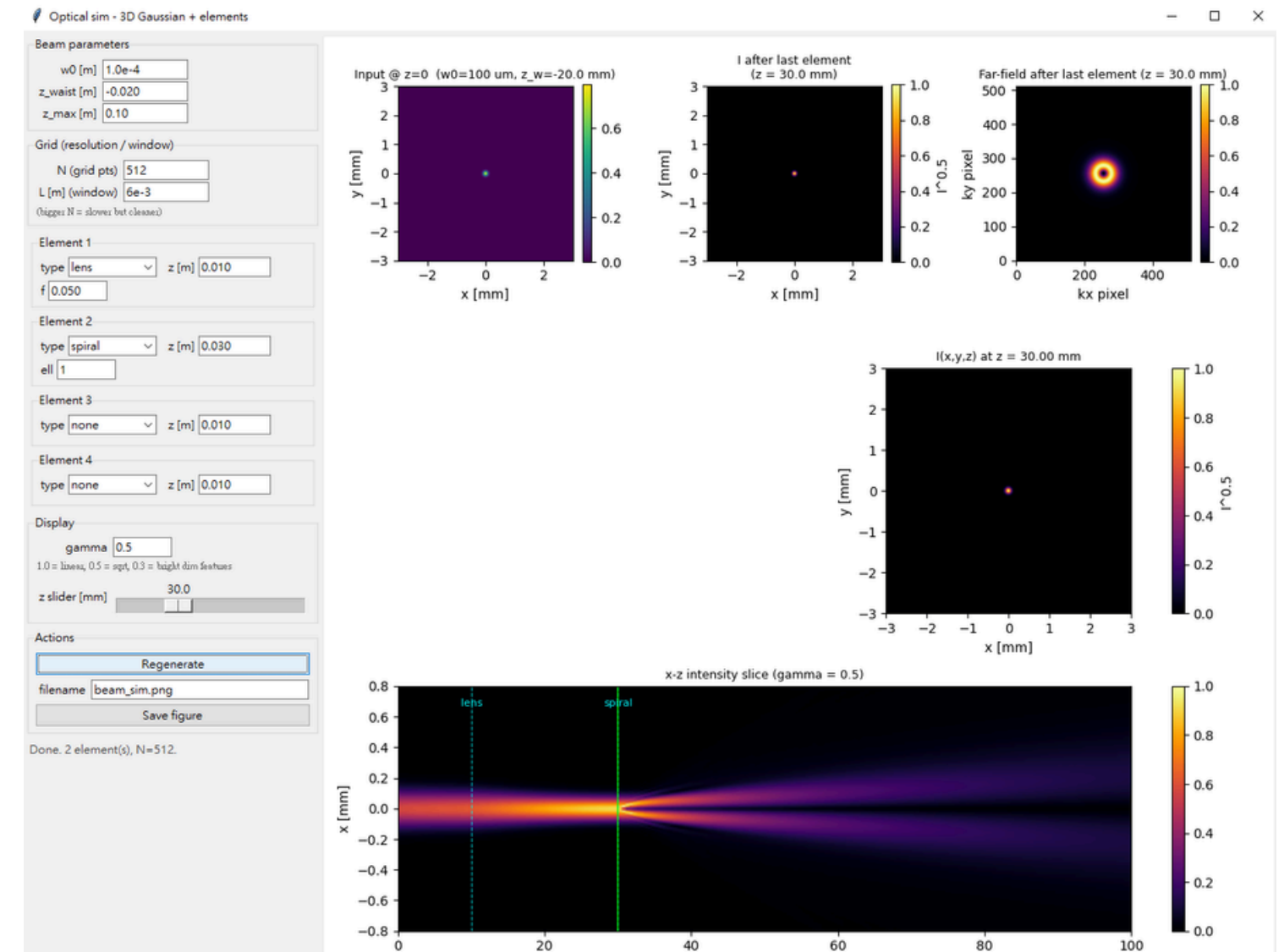
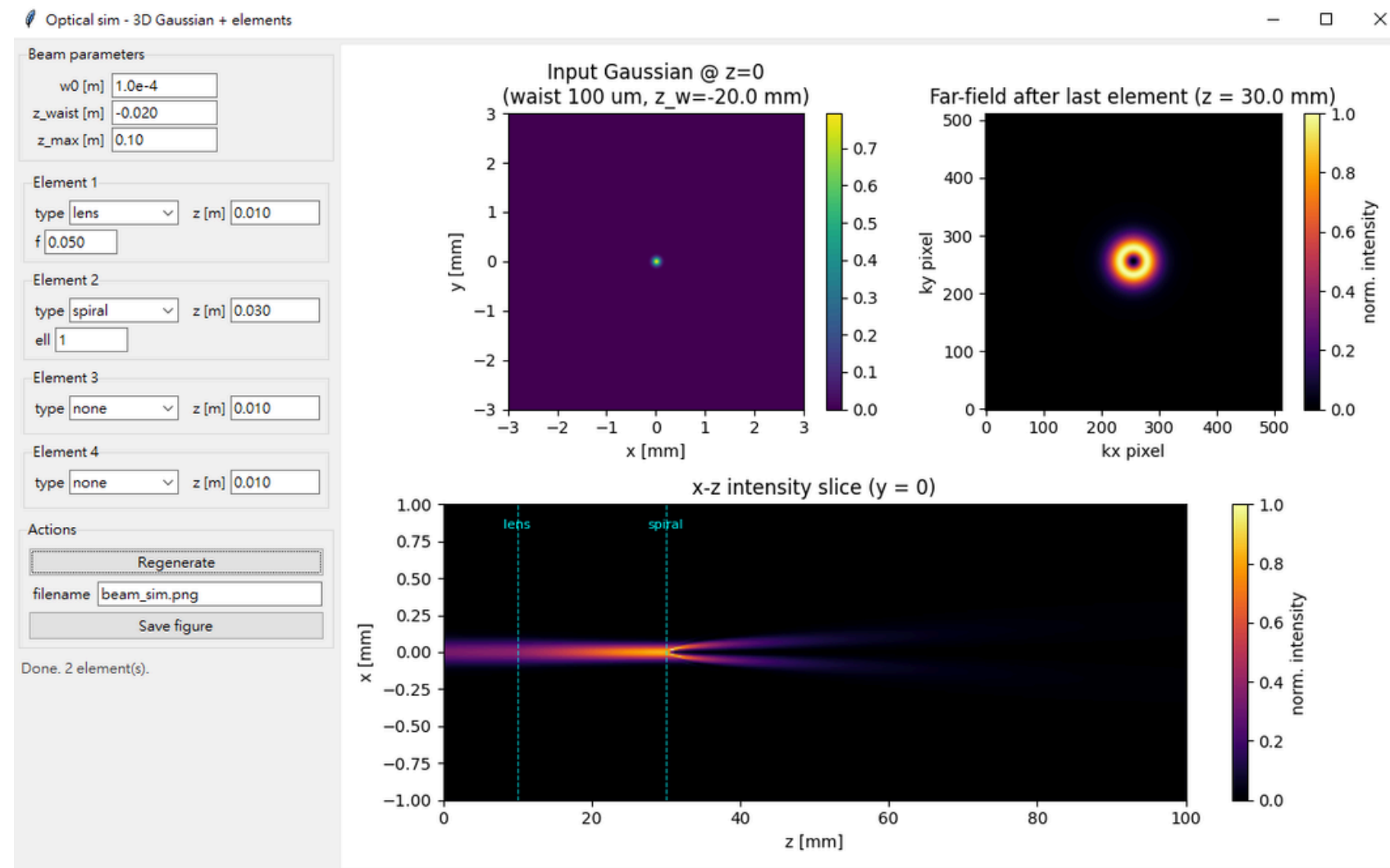
VERSION ITERATION



Version 0: only analysis single optical component

Version 1: A ray tracing GUI can support up to 4 optical components

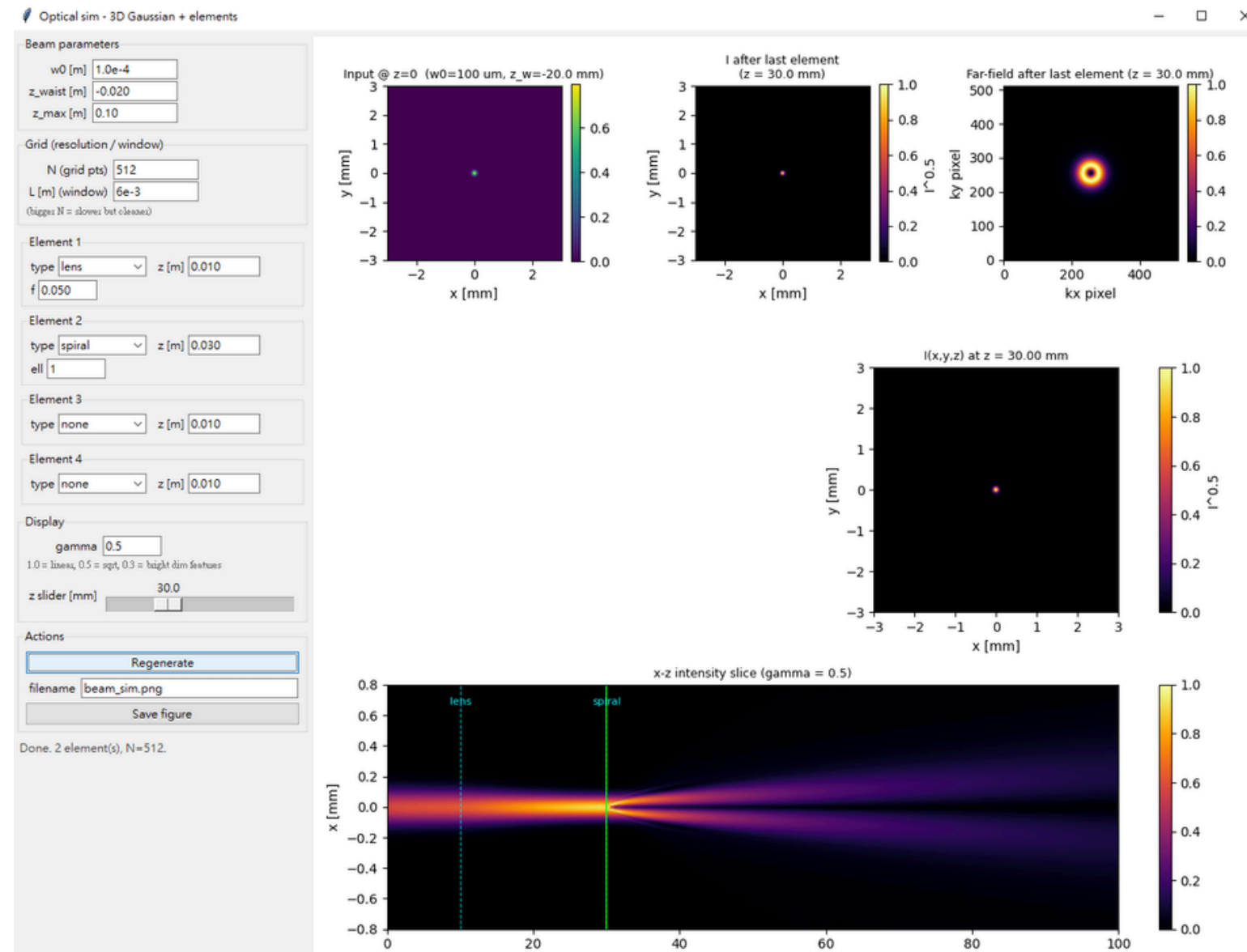
WORKFLOW



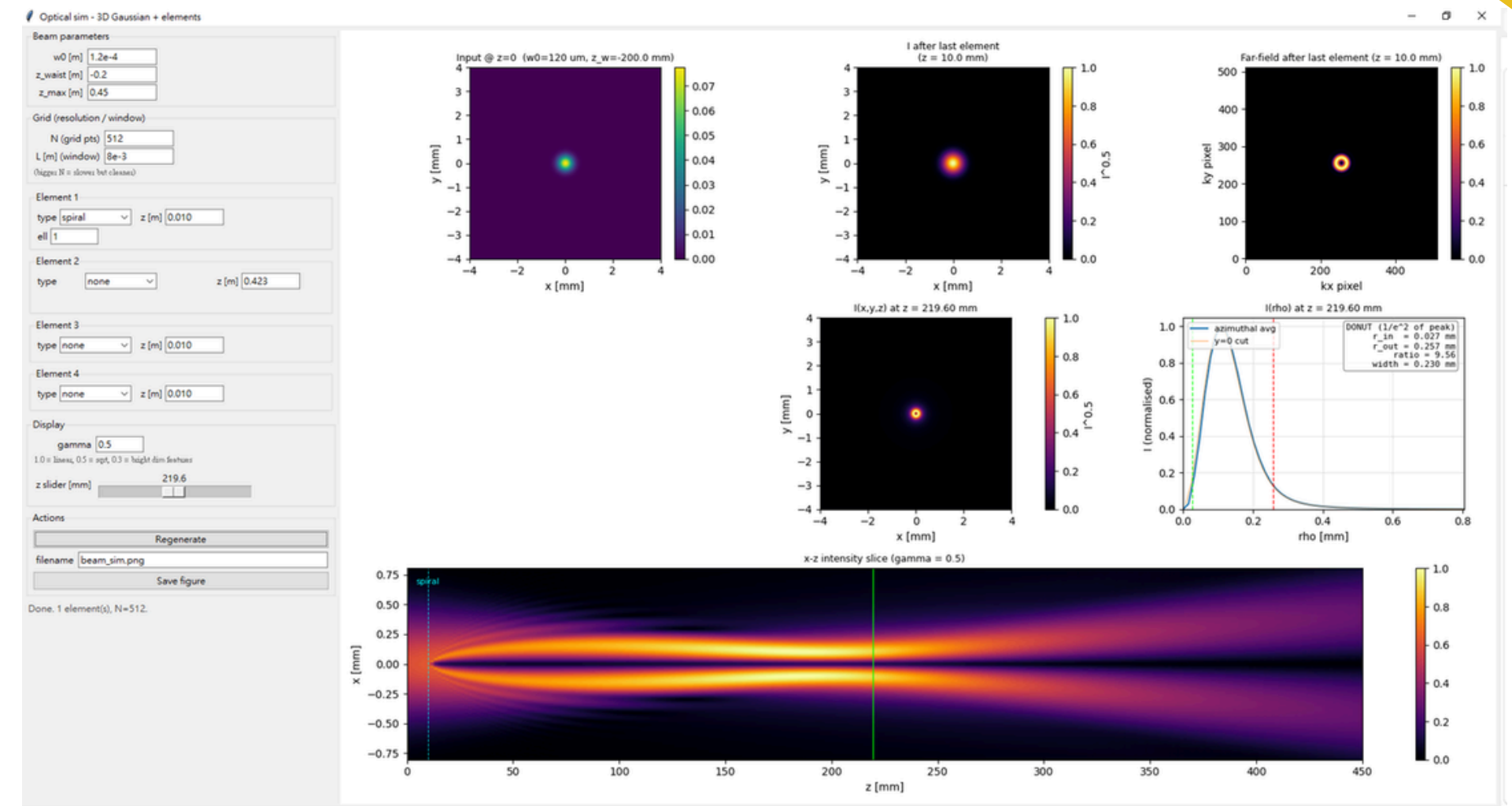
Version 1: A ray tracing GUI can support up to 4 optical components

Version 2: show the beam shape at different slice, provide beam shape right after the last component

WORKFLOW

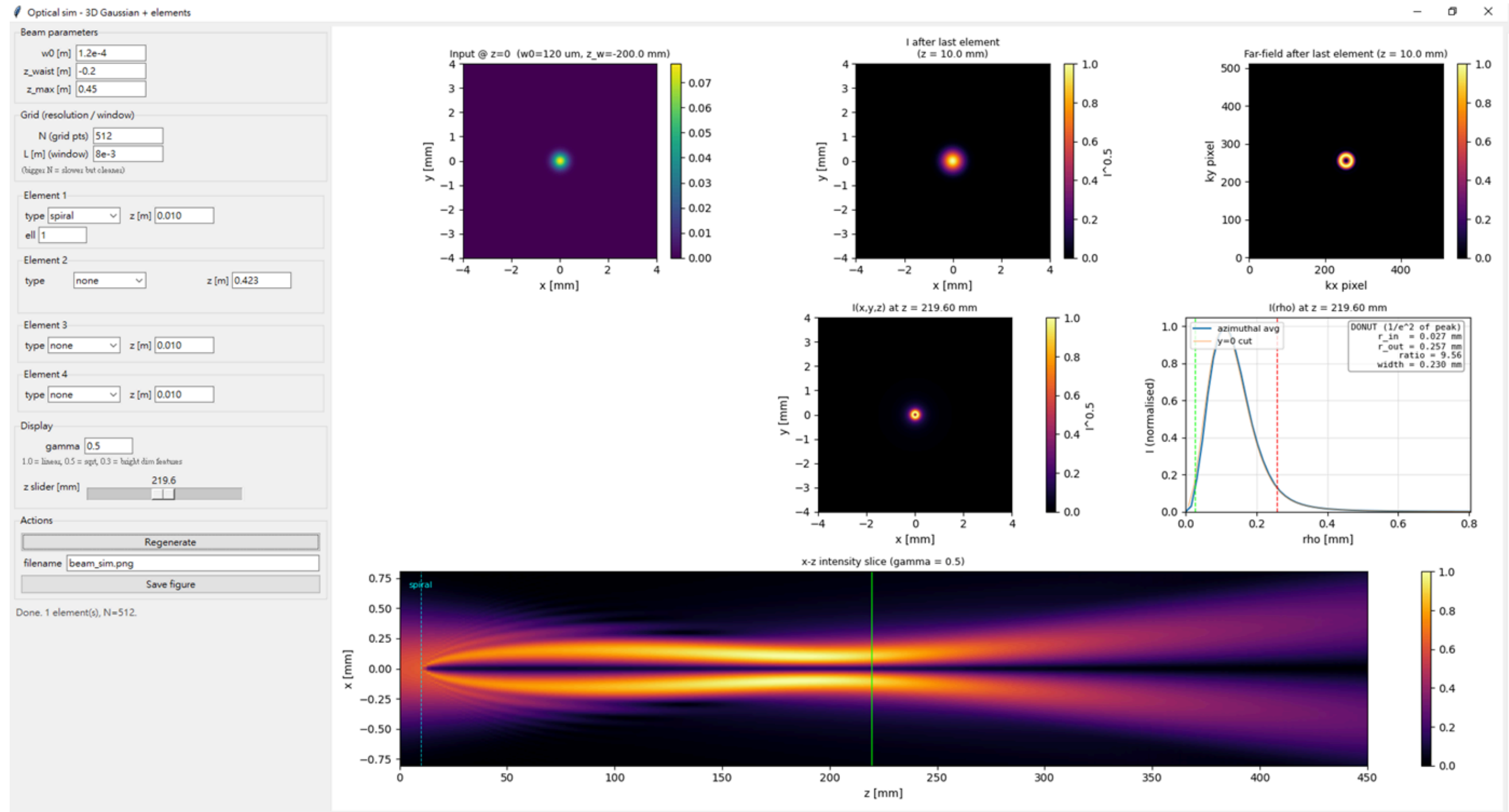


Version 2: show the beam shape at different slice,
provide beam shape right after the last component



Version 3: show the beam shape plot(I vs r)

RESULT



CHECK AND EVALUATION

Review the code line by line

- Understand the theory
- Checking it logic
- learn how to fix it
- Adding markdown

You need to Understand your code!

```
"""
GUI for the 3D Gaussian beam + optical-element simulator.

Features
-----
* 4 element slots: choose type (none / spiral / axicon / lens / aperture),
  | z position, and element-specific parameters.
* Global beam params: waist w0, z_waist, z_max.
* "Regenerate" recomputes the x-z slice with the current settings.
* "Save figure" writes the current plot to ./<filename>.png.

Run:  python optical_sim_gui.py
"""

import os
import numpy as np
import matplotlib
matplotlib.use("TkAgg")
import matplotlib.pyplot as plt
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
import tkinter as tk
from tkinter import ttk, messagebox

# =====
# Physics core (same propagator as before, wrapped as a function)
# =====
WAVELENGTH = 780e-9 # paper uses 0.9 um
NZ          = 250
k0 = 2 * np.pi / WAVELENGTH

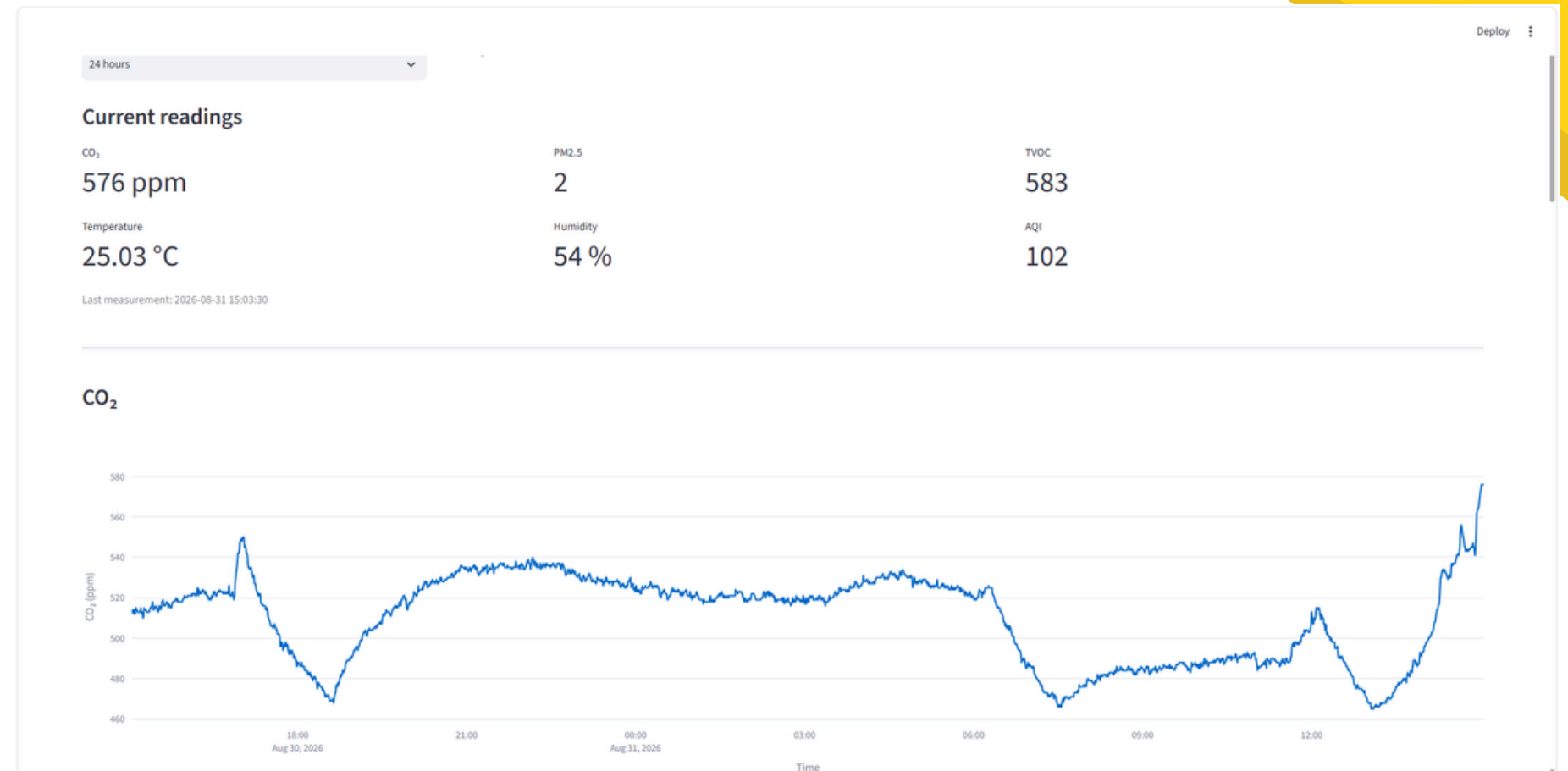
# Grid arrays – populated by _build_grids(N, L), called on every run
_x = _y = _dx = None
_X = _Y = _R = _Theta = None
_kz = None

def _build_grids(N, L):
    """(Re)build the spatial and frequency grids for given N and L."""
```

AI-GENERATED PROGRAM FOR AIR-QUALITY SENSOR

This sensor 8099-AP(blue-tooth) can monitor:

- Temperature, Humidity monitor
- CO2 concentration
- TVOC (***Total Volatile Organic Compounds***)
- PM2.5, PM10
- Give an Air Quality Index (AQI)



```
=====
Air Mentor 8099-AP
=====
CO2           : 777 ppm
PM2.5         : 1 ug/m3
PM10          : 2 ug/m3
TVOC          : 2357
Temperature   : 25.81 C
Humidity      : 50.0 %
AQI           : 183
RSSI          : -67 dBm
=====
CSV saved: 14:13:00 -> 2026-08-18.csv
```



MOTIVATION

- Product discounted in 2018
- App support stopped in 2018
- The company is no longer exist
- We only got a rice ball which can change colors



擎亞董事會通過子公司欣擎科技歇業事宜

Moneydj理財網
2018年1月24日



公開資訊觀測站重大訊息公告

(8096)擎亞-公告董事會通過子公司欣擎科技股份有限公司歇業事宜

- 1.事實發生日:107/01/23
- 2.公司名稱:欣擎科技股份有限公司
- 3.與公司關係(請輸入本公司或子公司):子公司
- 4.相互持股比例:不適用
- 5.發生緣由:欣擎科技股份有限公司乃本公司80%持股之子公司，考慮其營業成效及集團規劃經董事會決議通過歇業。
- 6.因應措施:無。
- 7.其他應敘明事項:無

MOTIVATION

**Can we find out the
blue tooth signal?**



WORKFLOW

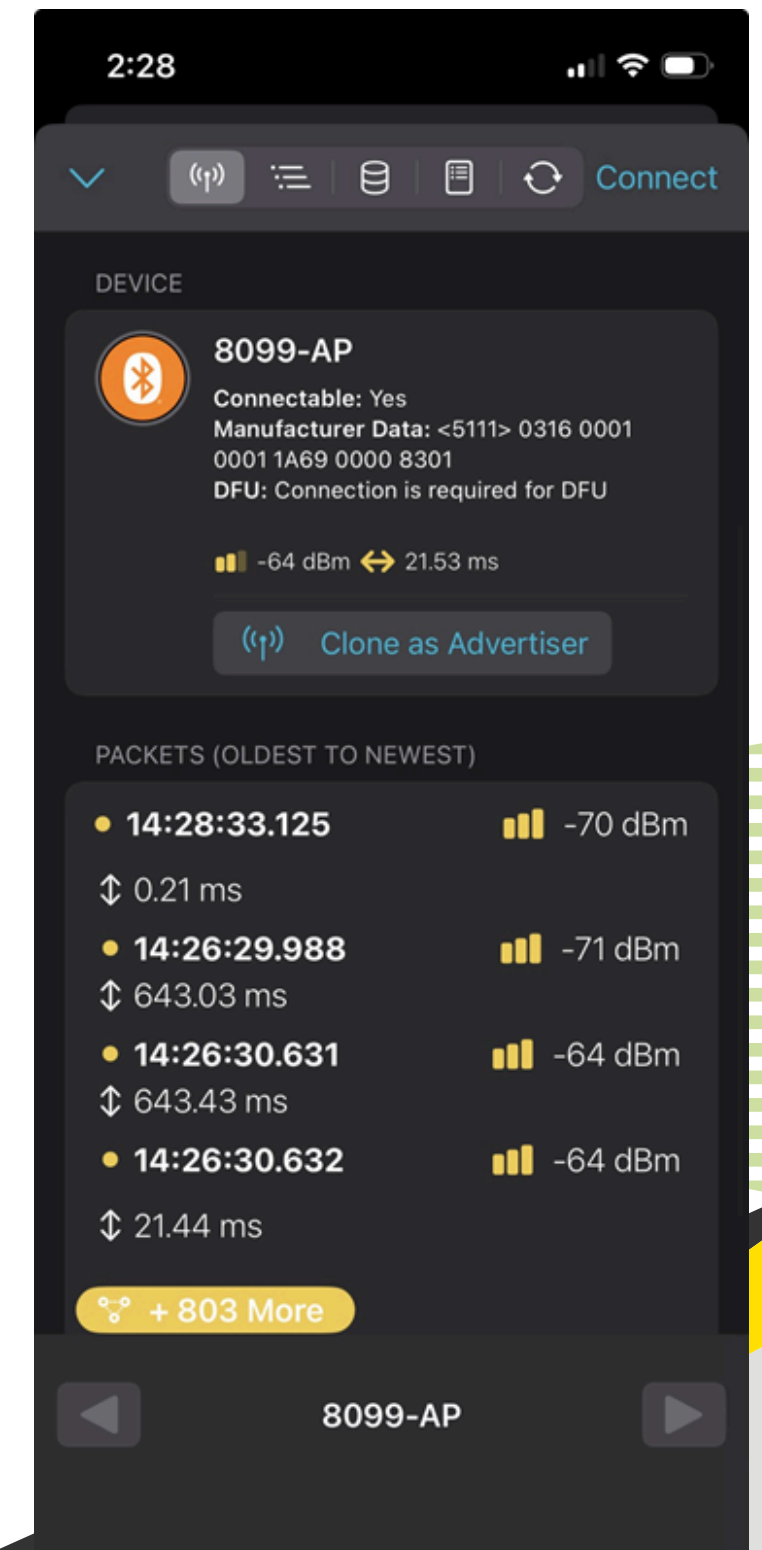
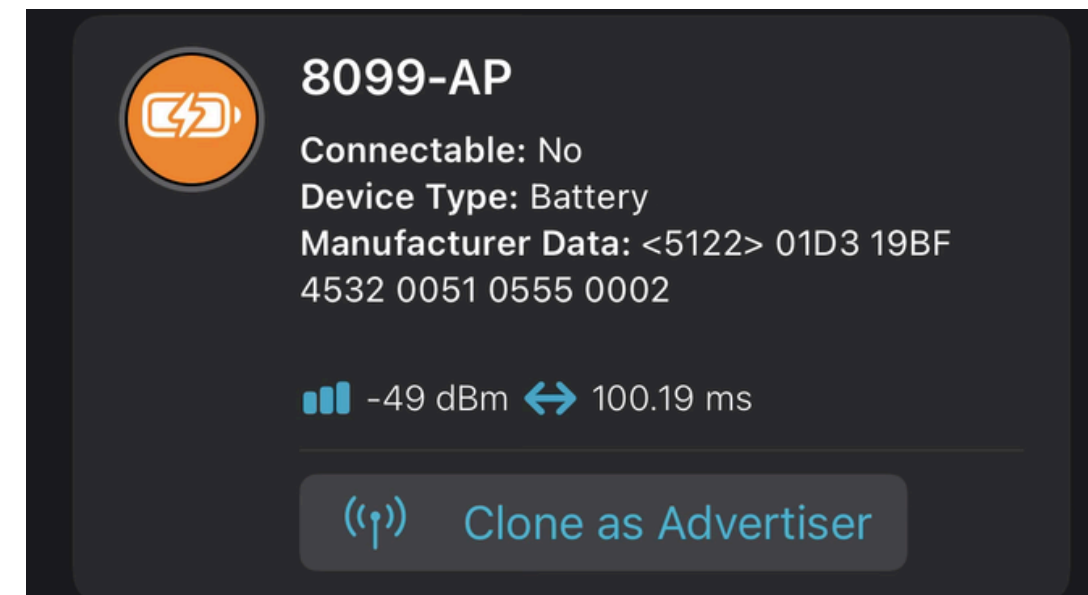


- Ask **chat GPT** can we check whether the device is working properly?

他們出的飯糰已經停產，監測用的app也在ios下架，我該怎麼重新使用這台儀器8099AP

- ChatGPT :Using **nRF Connect for Mobile**

- Paste the screen shot, then ChatGPT decode the received data.

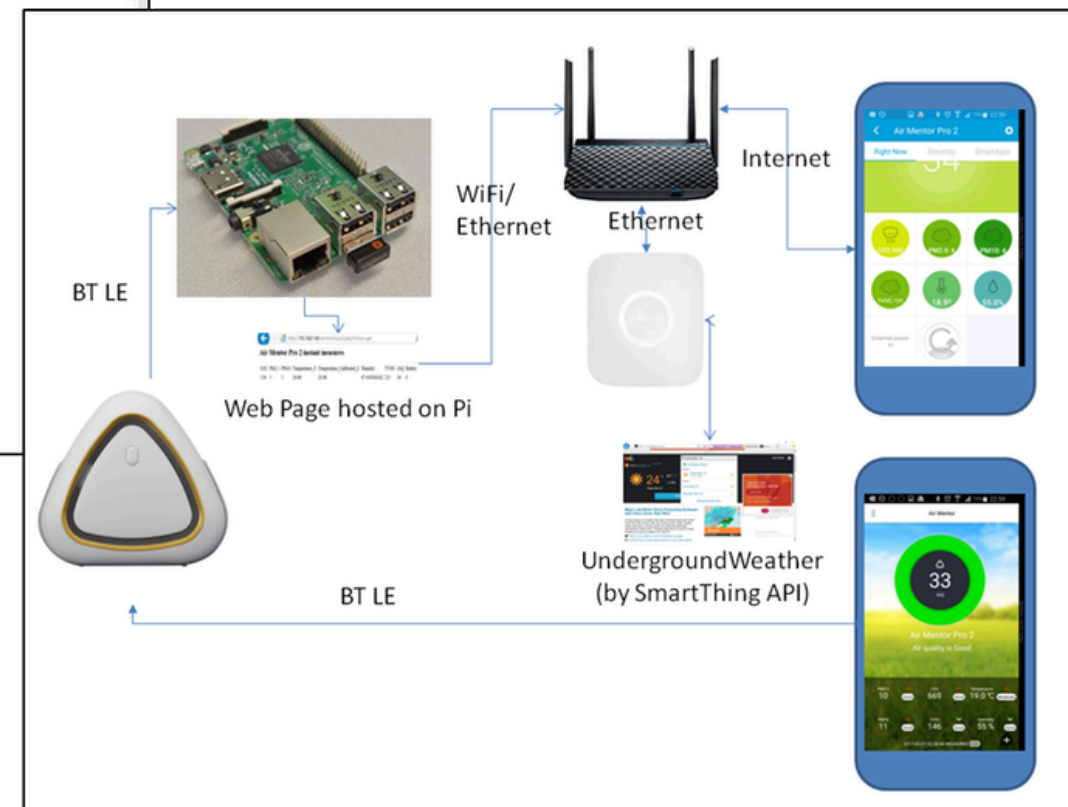
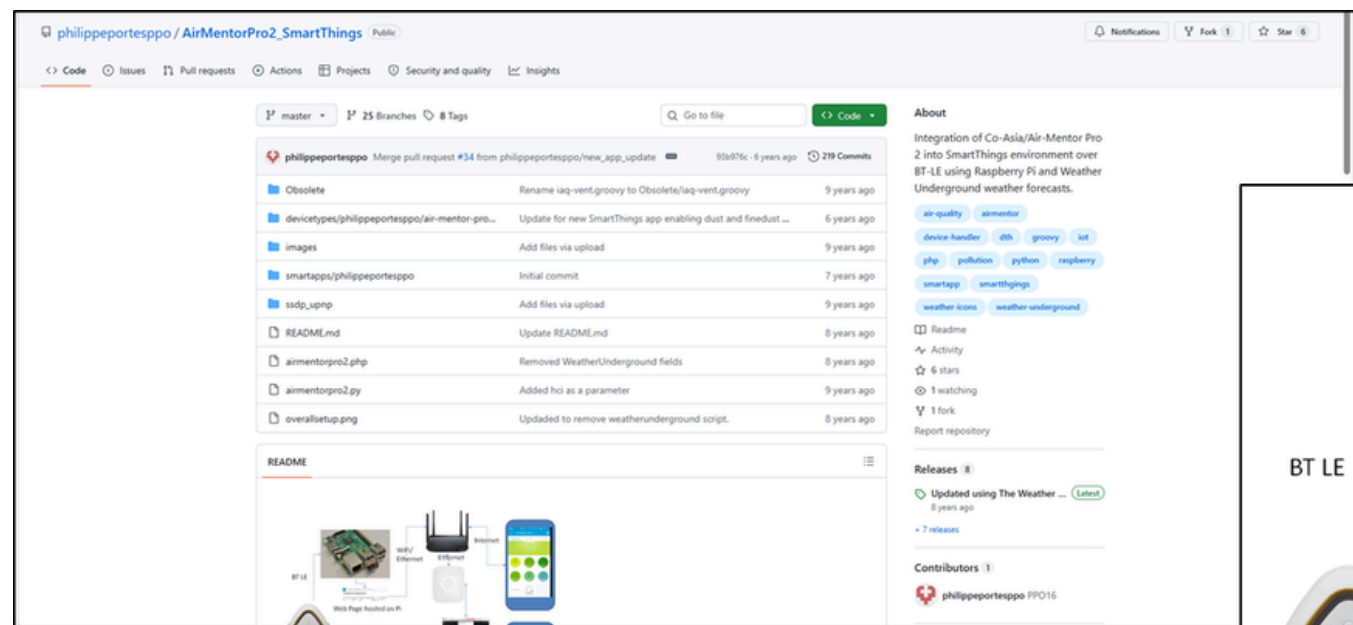


WORKFLOW



- Then my GPT decode the data into information (*by inference*)

I was suspecting(its inference) so I ask
it to search for github repository



```
Device name: 8099-AP
Connectable: No
RSSI: -49 dBm
Advertising interval: ~100 ms
```

```
Manufacturer Data:
<5122> 01D3 19BF 4532 0051 0555 0002
```

Air Mentor 8099-AP

Temperature	: 25.4 °C
Humidity	: 53 %
CO ₂	: 683 ppm
PM2.5	: 8 µg/m ³
PM10	: 11 µg/m ³
TVOC	: 0.13 ppm
RSSI	: -49 dBm

ref:philippeortesppo/AirMentorPro2_SmartThings: Integration of Co-Asia/Air-Mentor Pro 2 into SmartThings environment
over BT-LE using Raspberry Pi and Weather Underground weather forecasts.

WORKFLOW



- Ask ChatGPT to writing python code to monitor the received data.



```
Python
import asyncio
import math
import struct

from bleak import BleakScanner

# =====
# User settings
# =====

DEVICE_NAME = "8099-AP"

# =====
# Latest sensor values
# =====

latest = {
    "co2": None,
    "pm25": None,
    "pm10": None,
    "tvoc": None,
    "temperature": None,
    "temperature_calibrated": None,
    "humidity": None,
    "aqi": None,
    "rssi": None,
}
```



```
Found 8099-AP | RSSI = -72 dBm
Company ID: 0x5112 | Packet type: 0x12 | Payload: 06e119b9453100a905550002

--- Environment packet ---
TVOC      : 1761
Temp raw  : 0x1989
Temperature : 25.85 C
Temp delta : 69
Humidity   : 49.0 %
AQI        : 169
Extra      : 05550002

=====
Air Mentor 8099-AP
=====
CO2        : 806 ppm
PM2.5      : 1 ug/m3
PM10       : 1 ug/m3
TVOC       : 1761
Temperature : 25.85 C
Humidity    : 49.0 %
AQI        : 169
RSSI       : -72 dBm
=====
```

RESULT

Found 8099-AP | RSSI = -72 dBm
Company ID: 0x5112 | Packet type: 0x12 | Payload: 06e119b9453100a905550002

--- Environment packet ---

TVOC : 1761

Temp raw : 0x19B9

Temperature : 25.85 C

Temp delta : 69

Humidity : 49.0 %

AQI : 169

Extra : 05550002

=====

Air Mentor 8099-AP

=====

CO2 : 806 ppm

PM2.5 : 1 ug/m3

PM10 : 1 ug/m3

TVOC : 1761

Temperature : 25.85 C

Humidity : 49.0 %

AQI : 169

RSSI : -72 dBm

=====

Air Mentor Environmental Monitor

Environmental data recorded by Raspberry Pi 5

Time range

24 hours

Showing 2,881 measurements from 2026-08-30 15:45:00 to 2026-08-31 15:45:00

Current readings

CO₂

618 ppm

PM2.5

1

TVOC

538

Temperature

25.00 °C

Humidity

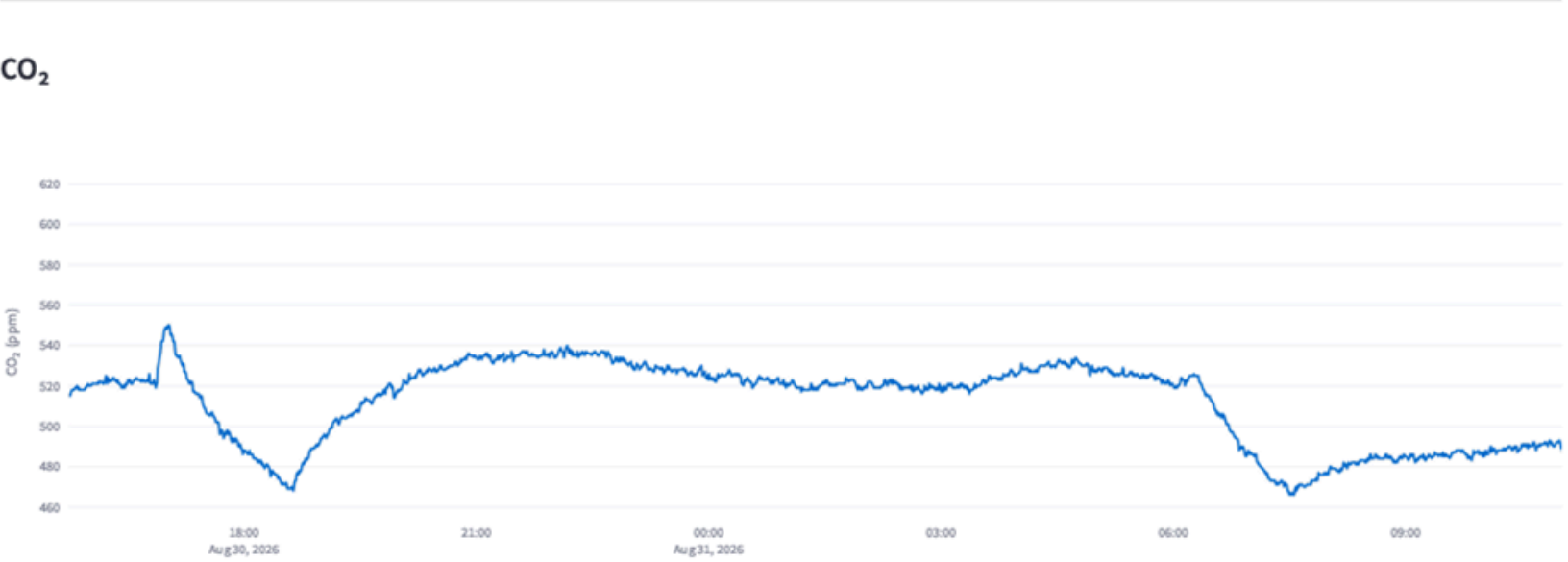
54 %

AQI

95

Last measurement: 2026-08-31 15:45:00

CO₂



CHECK AND EVALUATION

- Review the code line by line.
- Understand the code structure
- Calibration data with our old temperature and humidity sensor
- Check TVOC and CO2 by blow some breath to the device (we can see the number rise sharply)

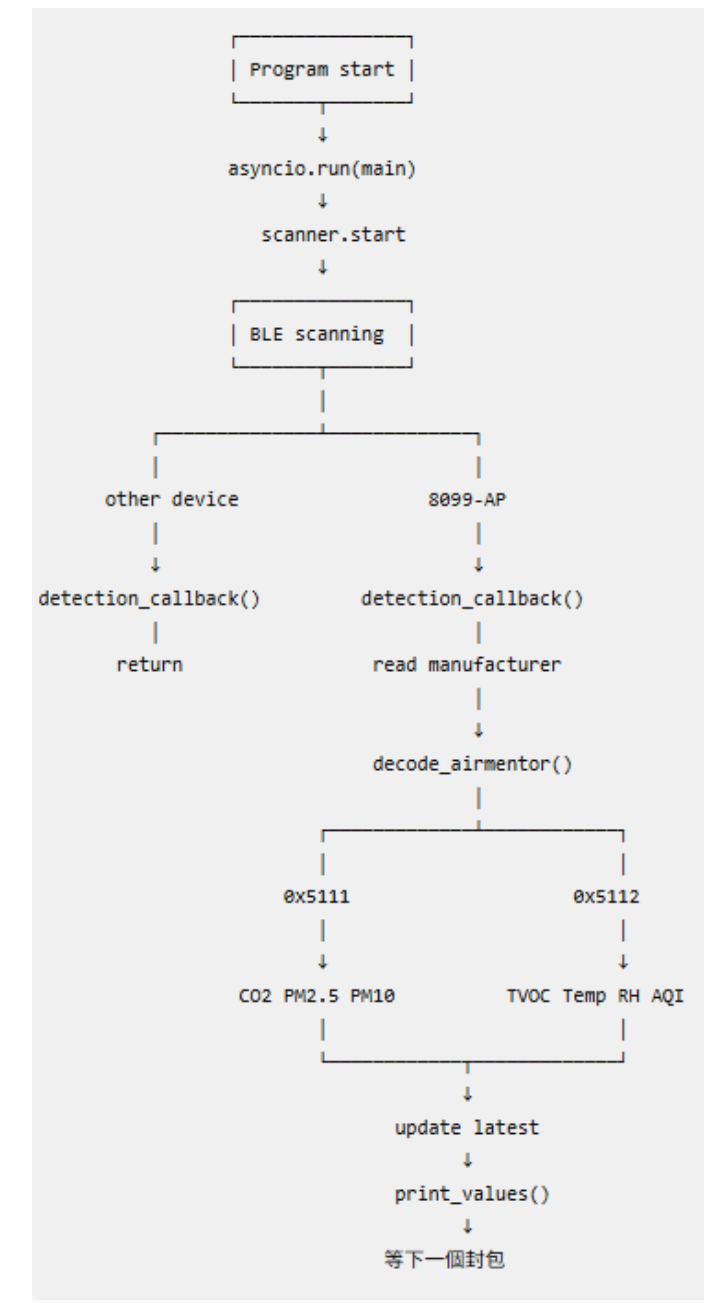
```
Found 8099-AP | RSSI = -67 dBm
Company ID: 0x5111 | Packet type: 0x11 |

--- CO2 / Particle packet ---
CO2   : 805 ppm ➡ up by 2000
PM2.5 : 1 ug/m3
PM10  : 1 ug/m3
Extra : 1a690000db01

=====
Air Mentor 8099-AP
=====

CO2           : 805 ppm
PM2.5         : 1 ug/m3
PM10          : 1 ug/m3
TVOC          : 1761
Temperature   : 25.85 C
Humidity      : 49.0 %
AQI           : 169
RSSI          : -67 dBm
=====
```

Code structure





Q & A

BACK-UP: ANGULAR SPECTRUM METHOD(ASM)

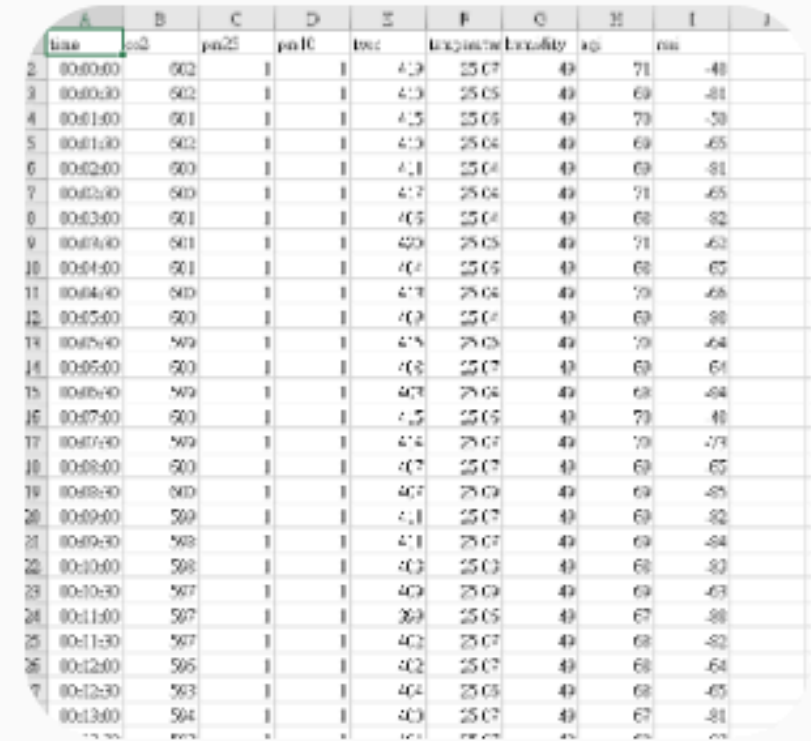
- $E(x, y, z) = FFT^{-1}(\bar{E}(k_x, k_y, z = 0) * H)$
- Where H is the transfer function or propagation transfer function:
- $H(k_x, k_y, z) = \exp(ik_z z)$

Since $k_z = \sqrt{k^2 - k_x^2 - k_y^2}$, $H = \exp(iz\sqrt{k^2 - k_x^2 - k_y^2})$

Ref: Joseph W. Goodman, **Introduction to Fourier Optics 2nd edition**

BACK-UP: PROMPT FOR BUILDING WEBSITE

- Tell the specific detail about what do you want to build to AI
- Better show the data strucutre

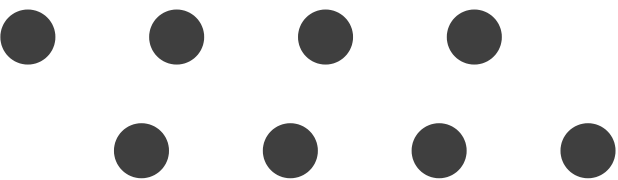


	A	B	C	D	E	F	G	H	I	J
	time	id	pm25	pm10	tsc	temperature	humidity	ap	pm	
2	00:00:00	502	1	1	4.3	25.07	49	71	-48	
3	00:00:30	502	1	1	4.3	25.05	49	69	-41	
4	00:01:00	501	1	1	4.5	25.05	49	79	-58	
5	00:01:30	502	1	1	4.3	25.04	49	69	-45	
6	00:02:00	500	1	1	4.1	25.04	49	69	-41	
7	00:02:30	500	1	1	4.2	25.04	49	71	-45	
8	00:03:00	501	1	1	4.5	25.04	49	68	-42	
9	00:03:30	501	1	1	4.0	25.05	49	71	-43	
10	00:04:00	501	1	1	4.0	25.05	49	68	-47	
11	00:04:30	500	1	1	4.1	25.04	49	70	-46	
12	00:05:00	500	1	1	4.0	25.04	49	69	-48	
13	00:05:30	500	1	1	4.1	25.05	49	70	-44	
14	00:06:00	500	1	1	4.0	25.07	49	69	-41	
15	00:06:30	500	1	1	4.0	25.04	49	68	-44	
16	00:07:00	500	1	1	4.5	25.05	49	79	-48	
17	00:07:30	500	1	1	4.4	25.07	49	70	-49	
18	00:08:00	500	1	1	4.7	25.07	49	69	-47	
19	00:08:30	500	1	1	4.0	25.09	49	69	-45	
20	00:09:00	500	1	1	4.1	25.07	49	69	-42	
21	00:09:30	508	1	1	4.1	25.07	49	69	-44	
22	00:10:00	508	1	1	4.0	25.09	49	68	-42	
23	00:10:30	507	1	1	4.0	25.09	49	69	-43	
24	00:11:00	507	1	1	3.9	25.05	49	67	-48	
25	00:11:30	507	1	1	4.0	25.07	49	68	-42	
26	00:12:00	506	1	1	4.0	25.07	49	68	-44	
27	00:12:30	503	1	1	4.4	25.05	49	68	-45	
28	00:13:00	504	1	1	4.0	25.07	49	67	-41	

如何建立一個網站，可以讀取電腦local端的csv file，取出裡面的資料繪製成長期趨勢圖。

每個csv代表不同的日期，內部則表示不同的時間，有辦法做成觀看長期趨勢的圖嗎(近三天)

Build a website that reads local CSV files and plots data trends from the past three days. Each CSV file represents one day and contains data recorded at different times.



THANK YOU

By Eric from 107

